



Summer Internship Project Report, 2023

Exploring the Nonlinear Dynamical Systems: A Study on Stability, Bifurcations, Chaos and Attractors



Arpan Saha

BSc Honours in Mathematics

Undergraduate 2nd Year

IVR No: 202100038997

Ramakrishna Mission Vidyamandira

Supervisor: Dr. Debaldev Jana

Ramakrishna Mission Vivekananda Educational and Research Institute

Acknowledgement

I would like to express my heartfelt gratitude and appreciation to those who have supported me throughout my project work in the area of Nonlinear Dynamics. This accomplishment would not have been possible without their guidance, encouragement, and unwavering support.

First and foremost, I would like to thank my supervisor, Dr. Debaldev Jana for his invaluable guidance, motivational talks, expertise, and continuous support. His mentorship has been instrumental in shaping my understanding of nonlinear dynamics and guiding me towards the right direction throughout this project. I am truly grateful for his patience, insightful feedback, and commitment to my academic growth. I would like to acknowledge his unwavering commitment to excellence, as he consistently challenged me to explore new perspectives and pushed me to surpass my own limitations. His insightful feedback and constructive criticism have been pivotal in growing my research skills and refining my project objectives.

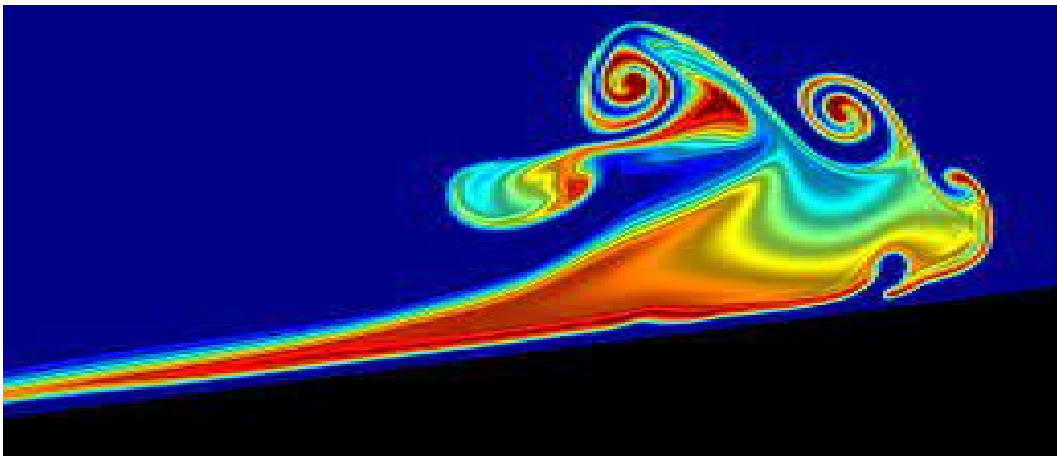
I would also like to extend my sincere appreciation to my teachers at Ramakrishna Mission Vidya-mandira who have helped me in my academic growth by fulfilling the prerequisite criteria of this project work. Their passion for teaching, engaging lectures, and willingness to address my queries have been integral to my learning experience. I am indebted to their commitment to academic excellence and their efforts in nurturing my intellectual development.

Furthermore, I would like to express my heartfelt gratitude to my family for their unwavering support and understanding. Their encouragement, love, and belief in my abilities have been a constant source of motivation for me in the entire project work.

Sincerely,
Arpan Saha

Title of Research Project

*Exploring the Nonlinear Dynamical Systems: A
Study on Stability, Bifurcations, Chaos and
Attractors*



Objectives

Nonlinear Dynamics is one of the most fascinating branches in Mathematics which deals with the behaviour of different Physical, Biological and many other different systems under certain specific conditions. Being a newcomer in this field, the primary objective of my project is to understand the different dynamical systems and then enter into much deeper contexts. The primary objectives of my projects are discussed in the following:

- 1. Analyzing the Nonlinear Differential Equations:** Studying the Nonlinear Dynamical Systems is solely based on analyzing the Nonlinear Differential Equations. Nonlinear Differential Equations are complicated and not so many methods are there to solve them, unlike Linear Differential Equations. The project aims to shed light on this context and predict their nature to study various Nonlinear Dynamical Systems. It is very important to note that the project confined its study to the Nonlinear Ordinary Differential Equations because of the lack of exposure of the Intern to Partial Differential Equations.
- 2. Understanding of Dynamical Systems:** To study the behaviour of different physical, and biological systems it is very important to have an understanding of what actually dynamical systems are. Then the different Nonlinear Phenomena like the behaviour, stability can be analysed.
- 3. Characterization of Stability Criteria and Bifurcations:** To study the various conditions under which the system will be stable. To identify the critical points and different parameters and among them the critical parameters upon which the stability of the system is dependent. One of the most important objectives of the project is to determine the change in the values of the parameters for which the system undergoes certain qualitative changes and thus heading towards Bifurcations.
- 4. Investigate the Chaos and Attractors:** To analyze the Chaotic Behaviour of the Dynamical Systems and different characterizations of Chaos such as dependence on initial conditions, Butterfly Effect, different onsets, Chaotic Attractors. The project aims to study the onset of Chaos in Dynamical Systems if any.
- 5. Mathematical Modelling:** To study different Physical, Chemical, Biological systems and investigate them mathematically and study different changes in them. The most important objective of the project is to analyze a Mathematical system by different methods.

Having these objectives the project on Nonlinear Dynamical Systems explains very much about different phenomena and thus seeks advancement in this topic contributing in the area of Complex Dynamics.

Contents

1	Introduction of Research Topic	5
2	Theoretical Framework of Topic	6
3	Profile of Organization	9
4	Methodology Followed	10
5	Analysis and Interpretation of Research Project	11
6	Conclusion & Suggestion of Research Project	26
7	Results Achieved	27
8	References	29

Chapter 1

Introduction of Research Topic

In today's life, the aesthetic appeal of Chaos and Fractals has created much enthusiasm even in the mind of non-mathematical people. But being a student of Mathematics background it is very usual to have the urge to go deeper. To have a glimpse of the immense beauty of these topics we have to go through a deeper field of study, Dynamics. Basically, it deals with the change in various systems over time. Although it is an interdisciplinary branch today, it is actually a branch of physics which started its journey in the 1600s when Sir Isaac Newton invented Calculus. Since then many Mathematicians and physicists have contributed to this subject.

The study of nonlinear dynamical systems has been very crucial for several decades for solving many complex Problems in different branches of Physics, Biology, Chemistry and Engineering. Though we have various methods for solving linear systems analytically, it is very difficult to solve Nonlinear systems analytically. Only very few Linear Approximations are available to change the systems into linear forms for this purpose. So this topic aims to be improved because Nonlinear systems are vital in the realm of various fields. This project explores the wonderful world of Nonlinear Dynamical systems, focusing on four key aspects: Stability, Bifurcations, Chaos, and Attractors. These fundamental concepts form the pillars upon which our understanding of complex dynamic behaviour is built. By exploring their interplay and intricate relationships and having a glimpse of their applications in different aspects we can gain deeper insights into the rich dynamics exhibited by a wide range of systems, from Ecological and Biological systems to Physical and Chemical systems.

Chapter 2

Theoretical Framework of Topic

Nonlinear Dynamics is one of the most diversified disciplines across the research study which helps to analyze many physical, biological, and engineering systems mathematically and helps to have a very good theoretical knowledge regarding them. It mainly focuses on studying the behaviour of complex systems, which cannot be studied by linearity or suitable linear approximations. The Theoretical Framework of Nonlinear Dynamical systems provides a very well-equipped machinery to uncover the patterns hidden in different Dynamical systems and helps to go deeper in terms of knowledge regarding Nonlinear Dynamical systems by analyzing Stability, Bifurcations, Chaos and Attractors.

Nonlinear Differential Equations: Nonlinear Differential Equations are the differential equations that are not linear in the unknown functions and their derivatives. The Non-linearity occurs when the dependent variables and their derivatives are combined in a nonlinear manner such as division, multiplication or through logarithmic or exponential functions. In this project, we have to work with Nonlinear Ordinary Differential Equations. Some examples of Nonlinear Ordinary Differential Equations are given below:

1. $\frac{dy}{dx} = x + x^2y + x^3y^2 + x^4y^3$.
2. $\frac{dy}{dx} = x^2y + x^4y^n$.
3. $m\frac{d^2y}{dx^2} = -kx - \lambda x^3$.
4. $m\frac{d^2r}{dt^2} = -\frac{d}{dr}\left(-\frac{k}{r} + \frac{1}{2}\frac{l^2}{mr^2}\right)$.(etc.)

Dynamical Systems: A system of n first-order differential equations in the space R^n is called a dynamical system of dimension n which determines the time behaviour of an evolutionary process. The evolutionary process may describe (i) a continuous-time process and (ii) a discrete-time process. The first can be described by differential equations and the second can be described by difference equations (Iterated maps). Dynamical systems whose equations of motion have no explicit dependence on time are called autonomous systems. They are called non-autonomous systems if they have an explicit dependence on time. Some examples of dynamical systems are given below:

1. The most common example of a dynamical system is the Newtonian system which depends on the position and velocity. It takes the form

$$\begin{aligned}\frac{dx}{dt} &= y, \\ \frac{dy}{dt} &= F(x, y).\end{aligned}$$

This system can be viewed as a dynamical system of dimension 2.

2. The Logistic Growth Model can be described as the differential equation of the form

$$\frac{dx}{dt} = rx\left(1 - \frac{x}{k}\right)$$

This can be viewed as a dynamical system of dimension 1.

Why Non-linearity: The systems of linear differential equations can be solved analytically very well since there are several methods to compute them but there is no such general rule for solving a system of non-linear differential equations. Sometimes they can be approximated to linear differential equations such as the equation of motion of a pendulum

$$\frac{d^2\theta}{dt^2} + \alpha \frac{d\theta}{dt} + \frac{g}{L} \sin\theta = 0, \text{ where } \alpha \text{ is the damping coefficient.}$$

Now the equation can be approximated to a linear form

$$\frac{d^2\theta}{dt^2} + \alpha \frac{d\theta}{dt} + \frac{g}{L} \theta = 0,$$

since $\sin\theta \approx \theta$ for very small θ . But here so many cases are excluded which hinders the authenticity of the analysis of the system. So we must develop a study to research and analyze the systems of nonlinear differential equations.

Visualizing the systems over time: To observe a dynamical system over time we need a collection of data taken from an authentic source. Also, we need to analyze them pictorially to have a very clear idea regarding them. For this, we need some special terms called Phase Space, Phase Portrait and Time series analysis. Phase Space is a multidimensional space which contains the complete behaviour of each state of a dynamical system. Each point in the space corresponds to a unique state of a dynamical system. Here we will work with the two-dimensional systems for which the Phase Space turns into a plane called Phase Plane. The trajectories of the systems corresponding to the spaces are called Phase portraits. The evolution of the systems over time is determined by plotting the different states of a system with respect to different times. This is known as the Time series of the dynamical systems.

Stability Theory: It is the most fundamental aspect of a dynamical system. Upon response to small changes i.e. perturbations in a system it either restores it and backs to its original state or it diverges from its trajectory and becomes unstable. For studying the stability of a system we use different methods such as Lyapunov Stability Criteria, Linearization, Eigenvalue Analysis etc. More details regarding this concept are discussed later in this project.

Bifurcation Theory: The characteristics of a dynamical system depends upon several parameters. For a small change in those parameters, considerable changes can happen in a system. The parameters are called critical parameters and this phenomenon is known as Bifurcation. In this project, we will study several types of Bifurcations such as Saddle-Node Bifurcations, Transcritical Bifurcations, Pitch-fork Bifurcations, and Hopf Bifurcations. More details regarding these topics and their applications are discussed later in this project.

Chaos Theory: No definition of the term *Chaos* is universally accepted yet, but almost everyone agreed that there are mainly three ingredients used in the working definition. :

Chaos is the phenomenon of occurrence of bounded non-periodic evolution in completely deterministic nonlinear dynamical systems with highly sensitive dependence on initial conditions. This is called deterministic chaos since the governing equations are deterministic.

1. Non-periodic behaviour means that there are trajectories which do not settle down to fixed points, periodic orbits, or quasi-periodic orbits as $t \rightarrow \infty$.
2. "Deterministic" means that the system has no random or noisy inputs or parameters. The irregular behaviour arises from the system's non-linearity, rather than from noisy driving forces.
3. "Sensitive dependence on initial conditions" means that nearby trajectories separate exponentially fast i.e. have a positive Lyapunov Exponent.

To describe the dramatic extreme sensitive dependence of chaotic solution of a system of differential equations Lorenz coined the term butterfly effect. In his own words, it reads as follows: "As small a perturbation as a butterfly fluttering its wings somewhere in the Amazons can in a few days time grow into a tornado in Texas". That is even a minute perturbation can cause realizable effects in a finite time under chaotic evolution.

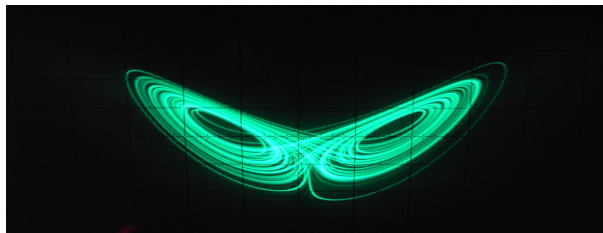


Figure: *Butterfly Effect*

Attractor Analysis: The term *attractor* is difficult to define in a rigorous way. We have to give such a definition which includes all its characteristics and excludes all other unuseful components.

Briefly describing an attractor is a set to which all the neighbouring trajectories converge. stable fixed points and stable limit cycles are good examples of this. To define more precisely, we define an **attractor** to be a closed set A with the following properties :

1. A is an invariant set: any trajectory $x(t)$ that starts in A stays in A for all time.
2. A attracts an open set of initial conditions: There is an open set U containing A such that if $x(0) \in U$, then distance from $x(t)$ to A tends to zero as $t \rightarrow \infty$. This means that A attracts all trajectories that start sufficiently close to it. The largest such U is called the *basin of attraction* of A .
3. A is minimal: There is no proper subset of A that satisfies conditions 1 and 2.

Chapter 3

Profile of Organization

Ramakrishna Mission Vivekananda Educational and Research Institute (RKMVERI) is a prestigious research institution with its roots deeply embedded in the name of Sri Ramakrishna and Swami Vivekananda. With its discipline and culture, it nurtures young talents from various fields both academically and spiritually.

Vision and Mission: Following Swamiji's vision on "Man Making" RKMVERI promotes the blending of both Eastern and Western Values respecting both the ancient Indian wisdom and work ethics, scientific mindset, research capability of the west. Besides building the scientific mindset among young talents, it also helps them learn social responsibilities to contribute to society by brain and heart. Albert Einstein's views as quoted below are very relevant in this context:

"It is essential that the student acquires an understanding of and a lively feeling for values. He must acquire a vivid sense of the beautiful and of the morally good. Otherwise, he with his specialized knowledge more closely resembles a well-trained dog than a harmoniously developed person."

Research Focus: Study of Nonlinear Dynamical systems is an important research area in modern days not only from a Mathematical perspective but also in the realm of various disciplines across Physics, Chemistry, Biology and Engineering. RKMVERI recognized the importance of research in this discipline and the framework of development in this area. RKMVERI has prominent research facilities along with the environment, peer research group, wonderful lab facilities and faculty members. RKMVERI organises seminars, workshops, conferences and various other activities to enhance the research work and motivation among the researchers. Collaborating with various academic institutions and research organizations the institute promotes exchange of knowledge and interdisciplinary research. Steven H. Strogatz's views are very precious in this context:

"Research is like a treasure hunt. You have a map but don't know where the treasure is hidden. So, you have to follow the clues and explore uncharted territory."

Chapter 4

Methodology Followed

To reach an important result via a project work it is important to follow several ways otherwise all the efforts would be in vain. The project work on Nonlinear Dynamical Systems comprises several methodologies to be pursued. The methodologies are described in the following:

Working on the Pillars: To pursue a project work it is mandatory to have a rigorous understanding of the basic concepts. To study Nonlinear Dynamical systems it is mandatory to have a good grasp of several topics like Real Analysis, Linear Algebra, Nonlinear Differential Equations etc.

Relevant Coursework: To explore the enriching world of a subject it is useful to have a glimpse of the subject. To study and explore Nonlinear Dynamical Systems one must have a comprehensive understanding of the subject as the first and foremost step. For this studying various books, going through several research papers, and attending relevant seminars, workshops, and conferences is very important because this builds the first connection to the subject.

Theoretical Analysis: After having a glimpse of the subject detailed theoretical analysis is very important. Studying different Mathematical models and the changes in them with time and analysing the stability properties, Bifurcations, Chaos, Attractors and different parameters especially the critical parameters is one of the main aspects of this project work.

Computational Techniques: Computational techniques play a very vital role in this project work. Since the project work is pursued over a very applied interdisciplinary field, practical knowledge along with theoretical knowledge is very much important in this regard. Computational simulations help to study the applied aspect of stability of a system and bifurcations, chaos and attractors. Along with these, they also help in visualising the different phenomena and observing the patterns in the systems that change over time.

Reviewing the Work: As a last step of the project work it is very important to review the data, computational results and theoretical study against the established theories so as to validate and verify the whole work correctly and finally aims to contribute in the advancement of knowledge in the field of Nonlinear Dynamics.

Chapter 5

Analysis and Interpretation of Research Project

1. Stability Analysis of Nonlinear Dynamical Systems:

• *Continuous Dynamical Systems:*

Consider a two-dimensional dynamical system,

$$\begin{aligned}\dot{x} &= P(x, y), \\ \dot{y} &= Q(x, y), \quad (\dot{} \equiv d/dt)\end{aligned}$$

Where P and Q have continuous first partial derivatives for all X and y and they does not depend explicitly on the variable t i.e. we are given an autonomous system here. Let

$$X = X^* = X_0 \equiv (x_0, y_0)$$

be the equilibrium point of the above system. Then at this point, the system does not change its state. So, $\dot{x} = \dot{y} = 0$, i.e. $P(x_0, y_0) = Q(x_0, y_0) = 0$. In order to determine stability, we slightly disturb it (or we say that we infinitesimally perturb it),

$$x = x_0 + \xi(t),$$

$$y = y_0 + \eta(t), \quad \text{where } \xi, \eta \ll 1$$

Now to determine the resultant motion, we use Taylor series expansion of the functions about the equilibrium point,

$$P(x, y) = P(x_0 + \xi, y_0 + \eta) = P(x_0, y_0) + \frac{\partial P}{\partial x}|_{x_0, y_0} \cdot \xi + \frac{\partial P}{\partial y}|_{x_0, y_0} \cdot \eta + \text{higher order terms in } (\xi, \eta)$$

$$Q(x, y) = Q(x_0 + \xi, y_0 + \eta) = Q(x_0, y_0) + \frac{\partial Q}{\partial x}|_{x_0, y_0} \cdot \xi + \frac{\partial Q}{\partial y}|_{x_0, y_0} \cdot \eta + \text{higher order terms in } (\xi, \eta)$$

Now using the fact that at the equilibrium point $P(x_0, y_0) = Q(x_0, y_0) = 0$, and neglecting the higher order terms (since they are tiny), the two-dimensional dynamical system can be rewritten as,

$$\begin{aligned}\dot{\xi} &= \dot{x} = a\xi + b\eta, \\ \dot{\eta} &= \dot{y} = c\xi + d\eta,\end{aligned}$$

where

$$a = \frac{\partial P}{\partial x}|_{x_0, y_0}, \quad b = \frac{\partial P}{\partial y}|_{x_0, y_0}$$

$$c = \frac{\partial Q}{\partial x}|_{x_0, y_0}, \quad d = \frac{\partial Q}{\partial y}|_{x_0, y_0}$$

Therefore the system evolves to the matrix of the form,

$$\begin{pmatrix} \dot{\xi} \\ \dot{\eta} \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix}$$

i.e. $\dot{X} = AX$

where $\dot{X} = \begin{pmatrix} \dot{\xi} \\ \dot{\eta} \end{pmatrix}$, $X = \begin{pmatrix} \xi \\ \eta \end{pmatrix}$ and $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is the **Jacobian** Matrix of the **Linearized** system.

Using standard methods the solution of the **Linearized** system is,

$$\xi(t) = A \exp(\lambda_1 t) + B \exp(\lambda_2 t)$$

$$\eta(t) = C \exp(\lambda_1 t) + D \exp(\lambda_2 t)$$

where A, B, C, D are constants and λ_1, λ_2 are eigen values of the matrix A .

Thus the stability nature of an equilibrium point is solely dependent on the eigen values of the matrix. Relation between the stability nature of the equilibrium point and the eigenvalues are described in the following:

- (a) **Stable:** Real parts of both the eigenvalues are negative.
- (b) **Unstable:** Real part of even one of the two eigenvalues is positive.
- (c) **Neutral:** Real parts of both the eigenvalues are zero.

Effect of The Small Nonlinear Terms: Is it really safe to omit the nonlinear terms of higher order in the process of Linearization? Is the qualitative nature of the Linearized system the same as that of the original system? The answer is yes. As long as the system is not like the cases when $\det A = 0$, the fixed equilibrium point is the same as predicted by the linearized system (See Andronov et al. (1973) for proof of this result). The borderline cases are much more delicate. They can be altered by small nonlinear terms.

Classification of Equilibrium Points:

- Case 1: $\lambda_1 \leq \lambda_2 < 0$ - Stable Node/Star

In this case the equilibrium point is stable. Thus any trajectory starting from the neighbourhood of the equilibrium point (x_0, y_0) in the $(x - y)$ phase plane approaches it exponentially in the long time limit. To study how the trajectories approach the equilibrium point we consider the slope dy/dx of the trajectories. Shifting the origin to the equilibrium point and rewriting the equations for new variables we get,

$$\frac{dy}{dx} = \frac{C\lambda_1 e^{(\lambda_1 - \lambda_2)t} + D\lambda_2}{A\lambda_1 e^{(\lambda_1 - \lambda_2)t} + B\lambda_2}$$

Now, we have to consider two cases.

(i) $\lambda_1 = \lambda_2$:

In this case, $\frac{dy}{dx} = \frac{C\lambda_1 + D\lambda_2}{A\lambda_1 + B\lambda_2} = m(\text{say})$. So it will become a straight line which passes through the equilibrium point. This type of equilibrium point is called **Stable Star**.

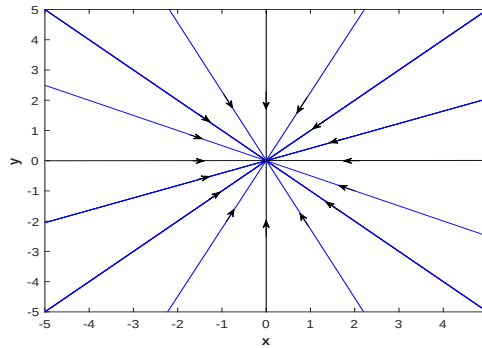


Figure: Stable Star

(ii) $\lambda_1 < \lambda_2$: Then the trajectories approach the equilibrium point in a parabolic path. This type of equilibrium point is called *Stable Node*.

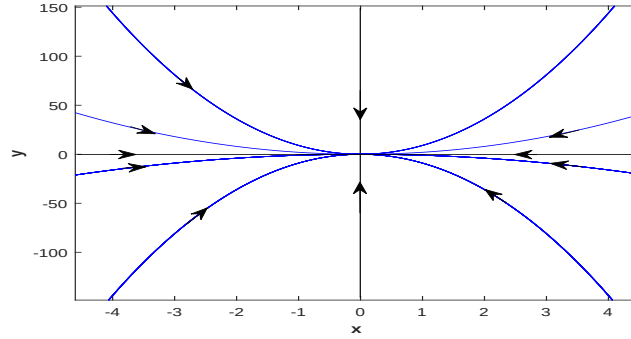


Figure: *Stable Node*

- Case 2: $\lambda_1 \geq \lambda_2 > 0$ - Unstable Node/Star

In this case trajectories diverge from the equilibrium point. For $\lambda_1 = \lambda_2$ the equilibrium point will be *Unstable Star*. For $\lambda_1 > \lambda_2$ the equilibrium point will be *Unstable Node*.

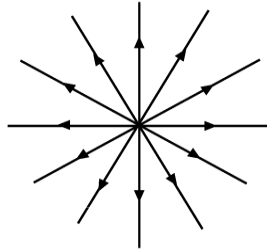


Figure: *Unstable Star*

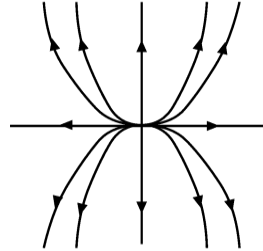


Figure: *Unstable Node*

- Case 3: λ_1, λ_2 complex conjugates - Stable/Unstable Focus

Let λ_1 and λ_2 are complex conjugates given by

$\lambda_1 = \alpha + i\beta$, $\lambda_2 = \alpha - i\beta$ where α, β are real constants and $\beta \neq 0$. Therefore the solution will be

$$\xi(t) = |A|e^{\alpha t} \cos(\beta t + \phi)$$

$$\eta(t) = |D|e^{\alpha t} \sin(\beta t + \phi_1)$$

Where

$$A = A_r + iA_i, D = D_r + iD_i$$

$$\phi = \tan^{-1}(A_i/A_r), \phi_1 = \tan^{-1}(D_i/D_r)$$

Now we consider two cases.

(i) $\alpha < 0$:

Both $|\xi|$ and $|\eta| \rightarrow 0$ as $t \rightarrow \infty$. Then the equilibrium point is stable. All the trajectories move to the point spirally before meeting with it asymptotically. Then the equilibrium point is called *Stable Focus*.

(ii) $\alpha > 0$:

The trajectories near to it moves away from it as $t \rightarrow \infty$. In that case the equilibrium point is called *Unstable Focus*.

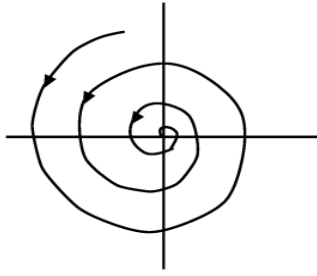


Figure: *Stable Focus*

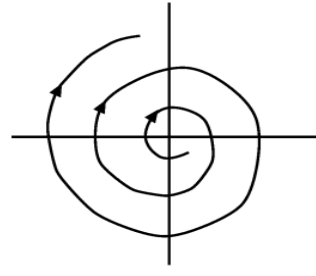


Figure: *Unstable Focus*

- Case 4: λ_1 and λ_2 pure imaginary and complex conjugates – Center/Elliptic

This is the case when $\alpha = 0$. Then $\lambda_{1,2} = \pm i\beta$. So,

$$\xi(t) = |A|\cos(\beta t + \phi)$$

$$\eta(t) = |D|\sin(\beta t + \phi_1).$$

Then the trajectories neither move towards it nor move away from it as $t \rightarrow \pm\infty$. This equilibrium point is called *Center type* or *Elliptic Equilibrium Point* and is neutrally stable. (To study its true stability properties we have to consider the neglected nonlinear terms.)

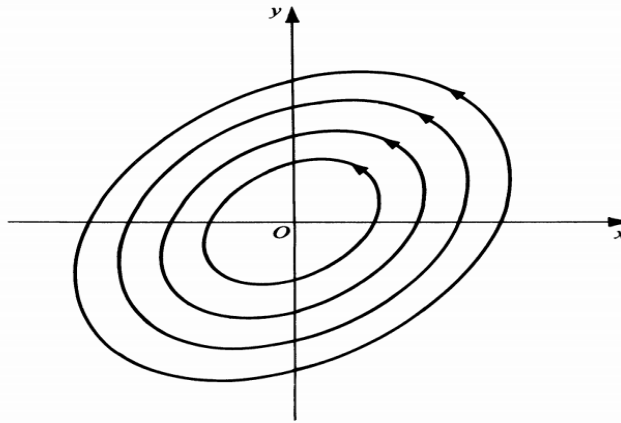


Figure: *Center*

- Case 5: $\lambda_1 < 0 < \lambda_2$ – Saddle Point

For any one of the eigenvalues less than zero and the other greater than zero one term will approach towards infinity and the other term approaches towards zero as $t \rightarrow \infty$. Overall, the point is unstable as the solutions diverge from it as time increases. This type of equilibrium point is known as *Saddle Point*. It is also called *Hyperbolic Equilibrium Point*.

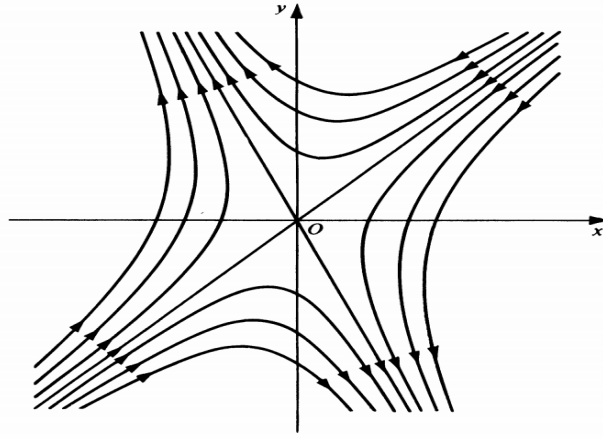


Figure: *Saddle Point*

- Case 6: Degenerate Cases (λ_1 or $\lambda_2 = 0$; both $\lambda_1 = \lambda_2 = 0$)

In this case, the dynamics of the system is governed by higher order terms in the Taylor Series Expansion. Such an analysis is beyond our scope.

From the above analysis we can appreciate that the stability of equilibrium points of Nonlinear Systems can be tested without actually solving the differential equation but by knowing the eigenvalues of the Jacobian matrix alone after performing suitable Linearization.

Limit Cycle Motion-The Closed Orbits:

In addition to the point attractors or repellers (i.e. the equilibrium points), two-dimensional dynamical systems can have isolated but closed orbits called limit cycles when certain conditions are satisfied.

Consider a two-dimensional dynamical system,

$$\begin{aligned}\frac{dx}{dt} &= P(x, y) \\ \frac{dy}{dt} &= Q(x, y)\end{aligned}$$

Where P and Q have continuous first partial derivatives for all x and y and they does not depend explicitly on the variable t i.e. it is an autonomous system.

Now we have a closed path C of the system which is approached spirally from either the inside or the outside by a non-closed path C_1 of the system either as $t \rightarrow \infty$ or as $t \rightarrow -\infty$. This closed path is called a limit cycle of the system.

- Example: Consider the two-dimensional dynamical system,

$$\begin{aligned}\frac{dx}{dt} &= y + x(1 - x^2 - y^2), \\ \frac{dy}{dt} &= -x + y(1 - x^2 - y^2).\end{aligned}$$

By rigorous calculations, the solution of the system becomes

$$\begin{aligned}x &= \frac{\cos t}{\sqrt{1+ce^{-2t}}}, \\ y &= -\frac{\sin t}{\sqrt{1+ce^{-2t}}}.\end{aligned}$$

The solutions of the system define the paths of the system in the xy plane. Examining these paths for various values of c , we note the following conclusions:

1. If $c = 0$, the path of the system is the circle $x^2 + y^2 = 1$, described in the clockwise direction.

2. If $c \neq 0$, then the paths will not be closed anymore, rather they will have a spiral behaviour.

(i) If $c > 0$, the paths are spiral lying inside the circle $x^2 + y^2 = 1$. As $t \rightarrow \infty$ they approach this circle.

(ii) If $c < 0$, the paths lie outside the circle $x^2 + y^2 = 1$. These outer paths also approach this circle as $t \rightarrow \infty$.

Since the closed path $x^2 + y^2 = 1$ is approached spirally from both inside and outside by non-closed paths as $t \rightarrow \infty$, we conclude that this circle is a limit cycle of the said system. The picture is shown in the following:

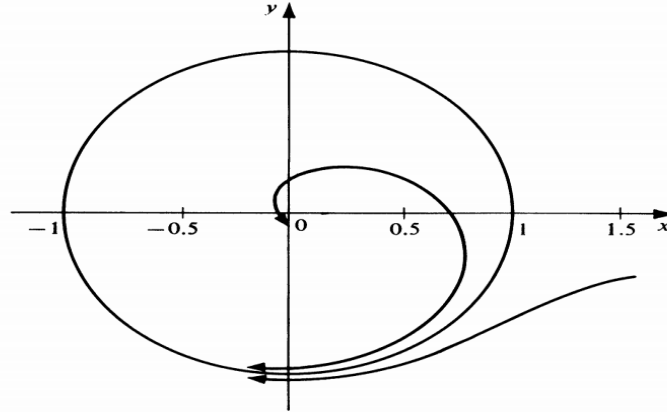


Figure: Limit Cycle

To study more about the closed orbit attractors or repellers, here about limit cycles we must assure about its existence. We state (without proof) two important theorems in the following regarding its existence:

- **Poincaré-Bendixson Theorem:**

Consider a two-dimensional dynamical system,

$$\begin{aligned}\dot{x} &= P(x, y), \\ \dot{y} &= Q(x, y), \quad (\dot{} \equiv d/dt)\end{aligned}$$

Where P and Q have continuous first partial derivatives for all X and y and they does not depend explicitly on the variable t i.e. we are given an autonomous system here.

If a solution curve $C : [x(t), y(t)]$ of the two-dimensional system (as described above) remains in a domain of the (x-y) phase space in which P and Q have continuous first partial derivatives for all $t \geq \tau$, for some τ , without approaching equilibrium points, then there exists a limit cycle in the domain, and either C is a limit cycle or it approaches a limit cycle as $t \rightarrow \infty$.

- **Bendixson's Nonexistence Theorem:**

Consider a two-dimensional dynamical system,

$$\begin{aligned}\dot{x} &= P(x, y), \\ \dot{y} &= Q(x, y), \quad (\dot{} \equiv d/dt)\end{aligned}$$

Where P and Q have continuous first partial derivatives for all X and y and they does not depend explicitly on the variable t i.e. we are given an autonomous system here.

For the above system, if $s = \frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y}$ does not change sign or vanish identically in a region D of the (x-y) phase space enclosed by a single closed curve, then no limit cycles lie entirely in D .

• **Discrete Dynamical Systems:**

By now we have seen the cases where the independent variable is continuous. There are several cases where the independent variable is discrete. This is the case in digital electronics, in parts of economics and finance theory, in impulsively driven systems and so on. These systems are often represented by difference equations or recursion relations or iterated maps, or simply maps. These maps could be of any dimension, depending on the number of physical variables. For example, the rule $x_{n+1} = \cos x_n$ is an example of a one-dimensional map, so-called because the points x_n belong to the one-dimensional space of real numbers and n takes only integer values. Generally for a map, $x_{n+1} = f(x_n)$, given an initial value x_0 , we compute x_1 , and then using x_1 compute x_2 and so on:

$$x_1 = f(x_0),$$

$$x_2 = f(x_1),$$

...

...

The sequence $x_0, x_1, \dots$ defines the trajectory of the variable x_n , where now the time n is discrete. Similarly one can consider higher dimensional maps.

Now, what is the nature of stability of the equilibrium points? To determine this, we have to perturb it slightly from its position and determine what happens next. Now,

We have the the map $x_{n+1} = f(x_n)$.

Let x^* be the equilibrium point.

Consider, $\delta_n = |x_n - x^*|$. Therefore,

$$\delta_{n+1} =$$

$$= |x_{n+1} - x^*|$$

$$= |f(x_n) - x^*|$$

$$= |f(x^* + \delta_n) - x^*|$$

$$= |x^* + \frac{df}{dx_n}|_{x_n=x^*} \delta_n + \dots - x^*| \quad (\text{By Taylor Expansion})$$

$$= |\frac{df}{dx_n}|_{x_n=x^*} \cdot \delta_n \quad (\text{Neglecting the higher order terms})$$

Therefore we have, $\delta_{n+1} = \lambda \delta_n$, where $\lambda = |\frac{df}{dx_n}|_{x_n=x^*}$. So, $\delta_1 = \lambda \delta_0$,

$$\delta_2 = \lambda \delta_1 = \lambda^2 \delta_0,$$

⋮

$$\delta_n = \lambda^n \delta_0,$$

Clearly, we see that $\delta_n \rightarrow 0$ only if $\lambda < 1$.

i.e. $|\frac{df}{dx_n}|_{x_n=x^*} < 1$. Thus the condition of stability of x^* is,

$$|\frac{df}{dx_n}|_{x_n=x^*} < 1$$

2. **Bifurcation Analysis of Nonlinear Dynamical Systems:**

• **Saddle-Node Bifurcation:**

Consider the two dimensional dynamical system,

$$\begin{aligned} \dot{x} &= \mu - x^2, \\ \dot{y} &= -y, \quad (\equiv d/dt) \end{aligned}$$

where μ is a parameter. We consider different cases here.

(i) $\mu < 0$:

The system has no real equilibrium points. The typical flow near the origin is illustrated in the following figure:

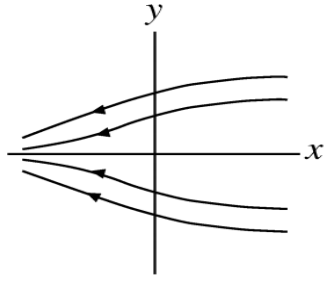


Figure: Phase Trajectories of the system for $\mu < 0$

(ii) $\mu = 0$:

There will be a unique equilibrium point and it will be the origin and the stability determining eigenvalues are $-1, 0$. So it is the non-hyperbolic case. The phase trajectories are as shown:

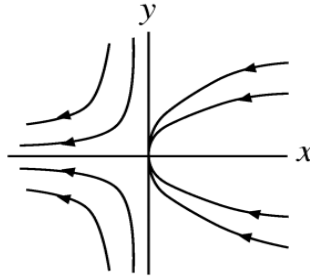


Figure: Phase Trajectories of the system for $\mu = 0$

(iii) $\mu > 0$:

Now we have two equilibrium points,

$$E_1: (x^*, y^*) = (+\sqrt{\mu}, 0),$$

$$E_2: (x^*, y^*) = (-\sqrt{\mu}, 0).$$

Following the previous articles, the stability determining eigenvalues for E_1 are $\lambda_1 = -1, \lambda_2 = -2\sqrt{\mu}$ and the stability determining eigenvalues for E_2 are $\lambda_1 = -1, \lambda_2 = 2\sqrt{\mu}$.

So from stability analysis, we can say clearly that E_1 is *Stable Node* and E_2 is *Saddle Point*. In this case, the phase trajectories are as shown:

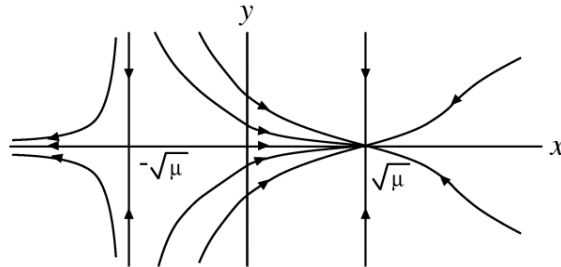


Figure: Phase Trajectories of the system for $\mu > 0$

Here we can see that for a negative value of μ there is no equilibrium point but when the value of μ starts approaching 0, we have one equilibrium point and when suddenly μ surpasses 0, we have two different equilibrium points. Thus for a small change of μ at 0, a bifurcation occurs giving birth to a Saddle Point and a Node. This type of bifurcation is known as *Saddle-Node Bifurcation*. Similarly, it can happen in arbitrary dimensions also. The associated *Bifurcation Diagram* (x versus μ) representing the qualitative change is as shown:

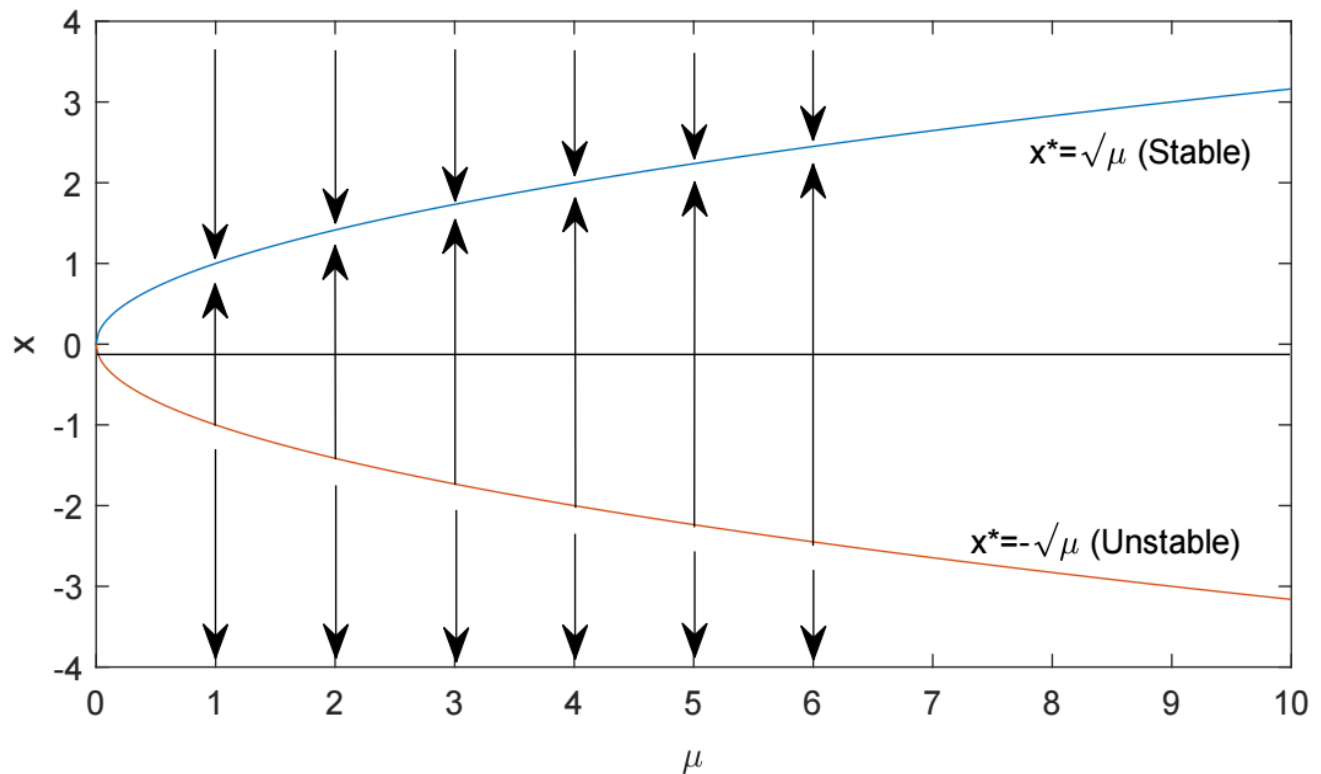


Figure: Saddle-Node Bifurcation diagram (Arrows indicating the Stable and Unstable Nature of the Equilibrium Points)

• **Transcritical Bifurcation:**

Consider the two-dimensional dynamical system,

$$\begin{aligned}\dot{x} &= -\mu x + x^2, \\ \dot{y} &= -y, \text{ Where } \mu \text{ is a parameter.}\end{aligned}$$

Now, we consider three cases.

(i) $\mu < 0$:

We have two equilibrium points,

$$E_0 : (x^*, y^*) = (0, 0),$$

$$E_1 : (x^*, y^*) = (\mu, 0).$$

And the eigenvalues,

corresponding to E_0 , $\lambda_1 = -1, \lambda_2 = -\mu$.

corresponding to E_1 , $\lambda_1 = -1, \lambda_2 = \mu$.

Therefore from Stability analysis, E_0 is Saddle Point and E_1 is Stable Node.

(ii) $\mu = 0$:

We have a unique equilibrium point, the origin i.e. $(0,0)$. The eigenvalues corresponding to it are $\lambda_1 = -1, \lambda_2 = 0$. So it is the Non-hyperbolic case.

(iii) $\mu > 0$:

We have two equilibrium points,

$$E_0 : (x^*, y^*) = (0, 0),$$

$$E_1 : (x^*, y^*) = (\mu, 0).$$

And the eigenvalues,

corresponding to E_0 , $\lambda_1 = -1, \lambda_2 = -\mu$.

corresponding to E_1 , $\lambda_1 = -1, \lambda_2 = \mu$.

Therefore from Stability analysis, E_0 is Stable Node and E_1 is Saddle Point. The Phase trajectories of the system corresponding to the three different conditions are given below:

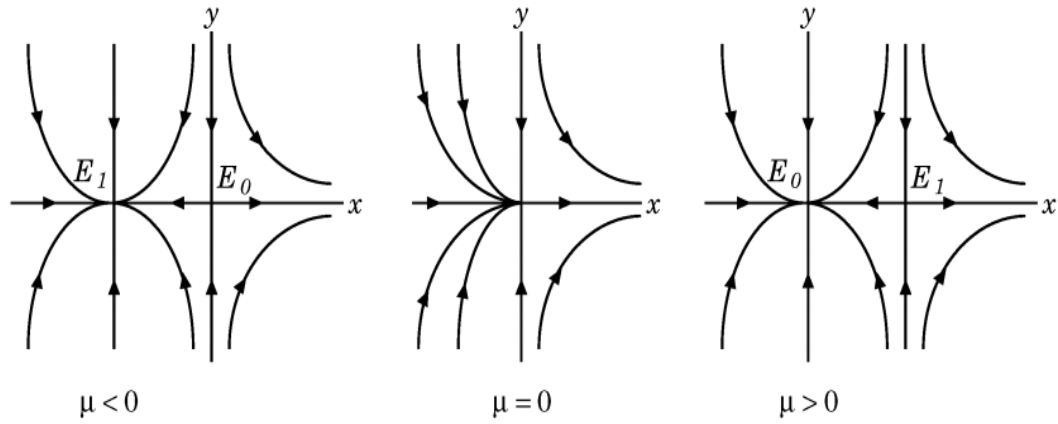


Figure: Phase Trajectories Corresponding to different values of μ

From the above analysis we see that the stability is exchanged (or transferred) at a critical value ($\mu = 0$) when the control parameter is varied. For this reason, the bifurcation is called transcritical bifurcation or exchange of stability bifurcation and is illustrated in the following diagram.

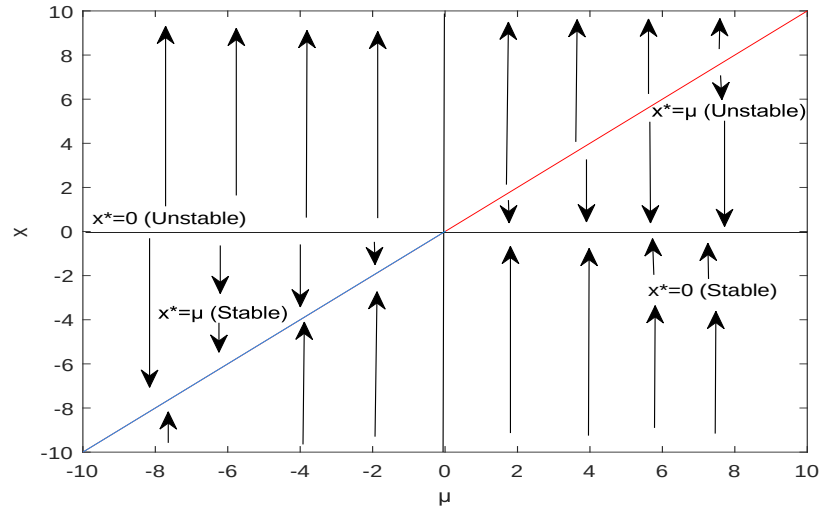


Figure: Transcritical Bifurcation diagram (Arrows indicating the Stable and Unstable Nature of the Equilibrium Points)

• **Pitchfork Bifurcation:**

Consider the dynamical system,

$$\begin{aligned}\dot{x} &= \mu x - x^3, \\ \dot{y} &= -y, \text{ Where } \mu \text{ is a parameter.}\end{aligned}$$

Now we consider different cases depending on the value of μ .

(i) $\mu < 0$:

Then also (0,0) is the only equilibrium point and the eigenvalues of the Jacobian Matrix corresponding to this

point are $\lambda_1 = -1, \lambda_2 = \mu$. So it is *Stable Node*. Therefore stability is changed.

(ii) $\mu = 0$:

Then only (0,0) is one equilibrium point and the eigenvalues of the Jacobian Matrix corresponding to this point are $\lambda_1 = -1, \lambda_2 = 0$. So it is the non-hyperbolic case.

(iii) $\mu > 0$:

Then we have three equilibrium points namely,

$$E_0 : (x^*, y^*) = (0, 0)$$

$$E_1 : (x^*, y^*) = (-\sqrt{\mu}, 0)$$

$$E_2 : (x^*, y^*) = (\sqrt{\mu}, 0)$$

Therefore, from stability analysis,

E_0 is Saddle Point.

E_1, E_2 are Stable Node.

The phase trajectories are shown below:

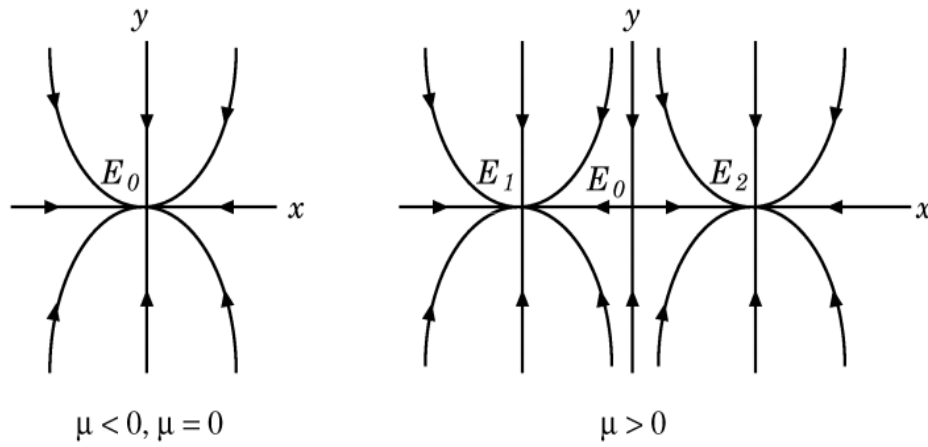


Figure: Phase Trajectories Corresponding to different values of μ

Therefore from the above analysis, we see as the parameter μ crosses the value 0, two new equilibrium points are born which are stable and the first equilibrium point loses its stability. Because of its tuning fork-like structure (depicted in the following diagram), it is known as *Pitchfork Bifurcation*.

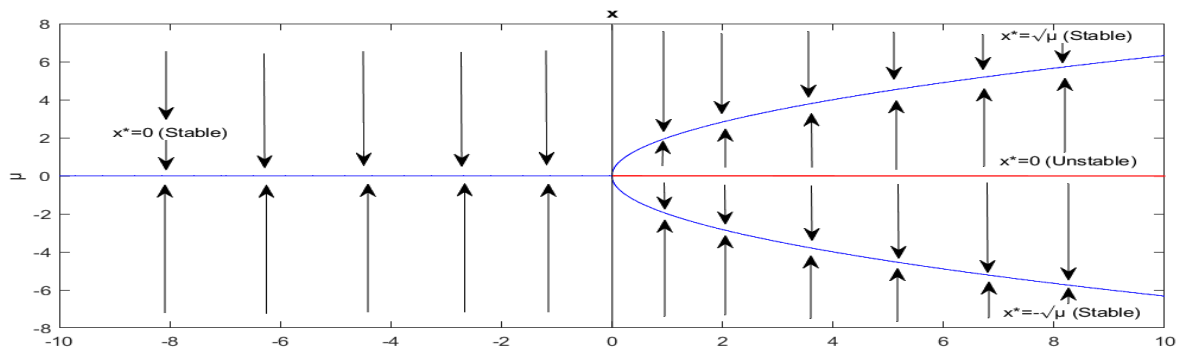


Figure: Pitchfork Bifurcation Diagram (Arrows indicating the Stable and Unstable Nature of Equilibrium Points)

• **Hopf Bifurcation:**

Consider the dynamical system,

$$\begin{aligned}\dot{x} &= y, \\ \dot{y} &= b(1 - x^2)y - x, \text{ Where } b \text{ is a parameter.}\end{aligned}$$

Clearly the equilibrium point is (0,0) and the corresponding eigenvalues of the Jacobian Matrix are $\lambda_{1,2} = \frac{b \pm \sqrt{b^2 - 4}}{2}$.

Now we consider different cases depending on the value of b .

(i) $-2 < b < 0$: The real part of the eigenvalues are negative. so (0,0) is a stable focus.

(ii) $b = 0$: The equilibrium point is now center and loses its stability.

(iii) $2 > b > 0$: The real part of the eigenvalues are positive and so the equilibrium point becomes the unstable focus.

Therefore from the above analysis, we see that the equilibrium point loses its stability as the parameter varies. To maintain the dynamical balance, the origin of a limit cycle happens. This type of bifurcation is known as *Hopf Bifurcation*. The phase trajectories are shown below:

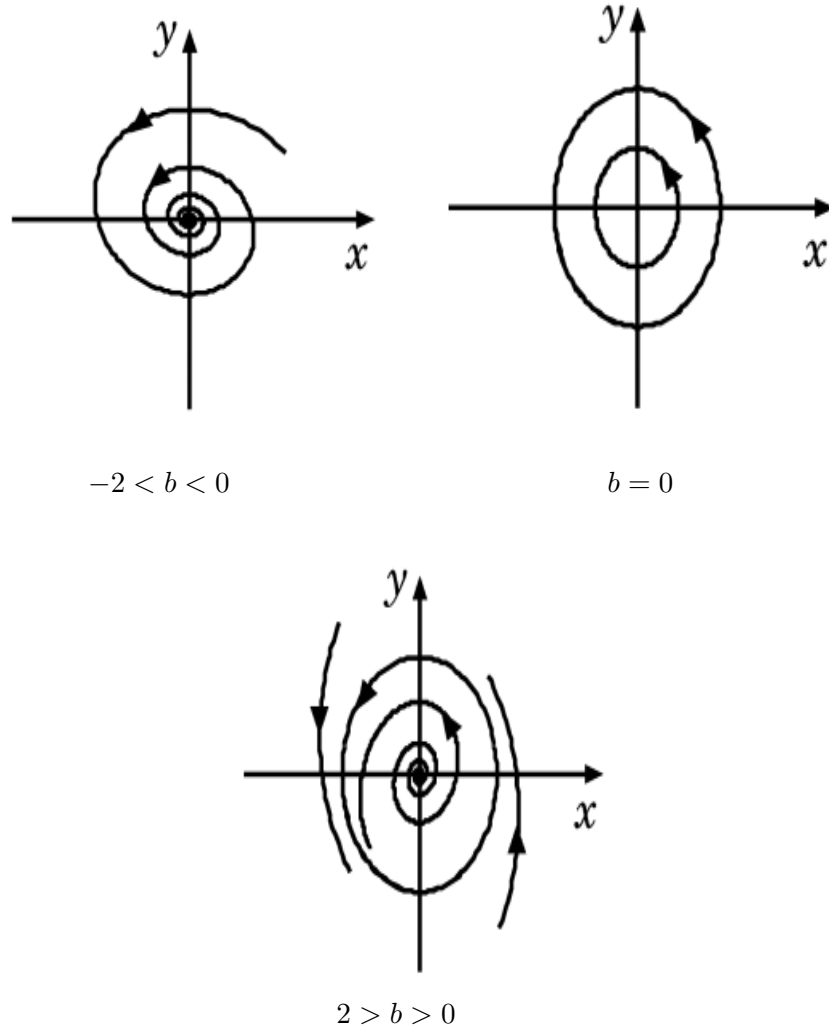


Figure: Phase Trajectories for different values of b

3. Mathematical Modelling:

• Theoretical Analysis:

Consider the two dimensional dynamical system,

$$\begin{aligned}\frac{dx}{dt} &= rx\left(1 - \frac{x}{k}\right) - \frac{\alpha(1-m)xy}{1+\alpha h(1-m)x} \\ \frac{dy}{dt} &= \frac{\theta\alpha(1-m)xy}{1+\alpha h(1-m)x} - \delta y\end{aligned}$$

where,

x, y = Prey, Predator Population Densities.

r = Intrinsic Growth Rate of Prey.

k = Environmental Carrying Capacity.

α = Foraging Efficiency or Attack Rate.

m = Prey Refuge Intensity, $0 < m < 1$.

θ = Food Conversion Efficiency.

h = Handling Time.

δ = Predator's Natural Death Rate.

Equilibrium Point	Feasibility Condition	Jacobian Matrix & Eigenvalues	Stability Result
$E_0 = (0, 0)$	Always	$J_{E_0} = \begin{pmatrix} r & 0 \\ 0 & -\delta \end{pmatrix}$, $\lambda_1 = r > 0$, $\lambda_2 = -\delta < 0$.	Saddle Point (Always)
$E_1 = (k, 0)$	Always	$J_{E_1} = \begin{pmatrix} -r & 0 \\ 0 & \frac{\theta\alpha(1-m)k}{1+\alpha h(1-m)k} - \delta \end{pmatrix}$ $\lambda_1 = -r < 0$, $\lambda_2 = \frac{\theta\alpha(1-m)k}{1+\alpha h(1-m)k} - \delta$.	(i) If $\delta > \frac{\theta\alpha(1-m)k}{1+\alpha h(1-m)k}$, E_1 is Stable Node. (ii) If $\delta = \frac{\theta\alpha(1-m)k}{1+\alpha h(1-m)k}$, E_1 is Non-hyperbolic Case. (iii) If $\delta < \frac{\theta\alpha(1-m)k}{1+\alpha h(1-m)k}$, E_1 is Saddle Point.
$E_2 = \left(\frac{\delta}{\alpha(1-m)(\theta-\delta h)}, \frac{r(k-x^*)\theta}{\alpha k(1-m)(\theta-h\delta)}\right)$ Where, $x^* = \frac{\delta}{\alpha(1-m)(\theta-\delta h)}$.	(i) $\theta > h\delta$. (ii) $m < 1 - \frac{\delta}{\alpha k(\theta-\delta h)}$.	$J_{E_2} = \begin{pmatrix} e & f \\ g & 0 \end{pmatrix}$ Where, $e = \frac{hr\delta}{\theta} \left[1 - \frac{\delta}{k\alpha(1-m)(\theta-\delta h)}\right] - \frac{r\delta}{k\alpha(1-m)(\theta-\delta h)}$, $f = \frac{r}{k} \left[\frac{\delta}{\alpha k(1-m)(\theta-\delta h)} - 1\right]$, $g = r(\theta-\delta h) \left[1 - \frac{\delta}{\alpha k(1-m)(\theta-\delta h)}\right]$, $\lambda_1 = \frac{e + \sqrt{e^2 - 4fh}}{2}$. $\lambda_2 = \frac{e - \sqrt{e^2 - 4fh}}{2}$.	Consider the case when $e^2 - 4fh < 0$. (i) If $m > 1 - \frac{\theta+h\delta}{hk\alpha(\theta-\delta h)}$, E_2 is Stable Focus. (ii) If $m = 1 - \frac{\theta+h\delta}{hk\alpha(\theta-\delta h)}$, E_2 is Center. (iii) If $m < 1 - \frac{\theta+h\delta}{hk\alpha(\theta-\delta h)}$, E_2 is Unstable Focus. At $m = 1 - \frac{\theta+h\delta}{hk\alpha(\theta-\delta h)}$ a limit cycle is born. Hence, <i>Hopf Bifurcation</i> Occurs in this case.

Table 1: Theoretical Analysis of the Two-dimensional Dynamical System

• Numerical Simulations:

To justify the theoretical model, we should have to consider the importance of experimental data. In the following, we consider several values assigned to the parameters to study the system numerically.

Parameter	Description	Assigned Value
x	Prey Density	Variable
y	Predator Density	Variable
r	Intrinsic Growth Rate of Prey	2.65
k	Environmental Carrying Capacity	898
α	Foraging Efficiency/Attack Rate	0.045
m	Prey Refuges Intensity	$0 < m < 1$
θ	Food Conversion Efficiency	0.215
h	Handling Time	0.0437
δ	Predator's Natural Death Rate	1.06

Table 2: Description and Assigned Values of the Parameters

The above parameter values show that the equilibrium point E_2 is stable when $0.1227 < m < 0.8445$. In this range, the point will be a *Stable Focus*. The corresponding phase portrait and time series of the system are shown below:

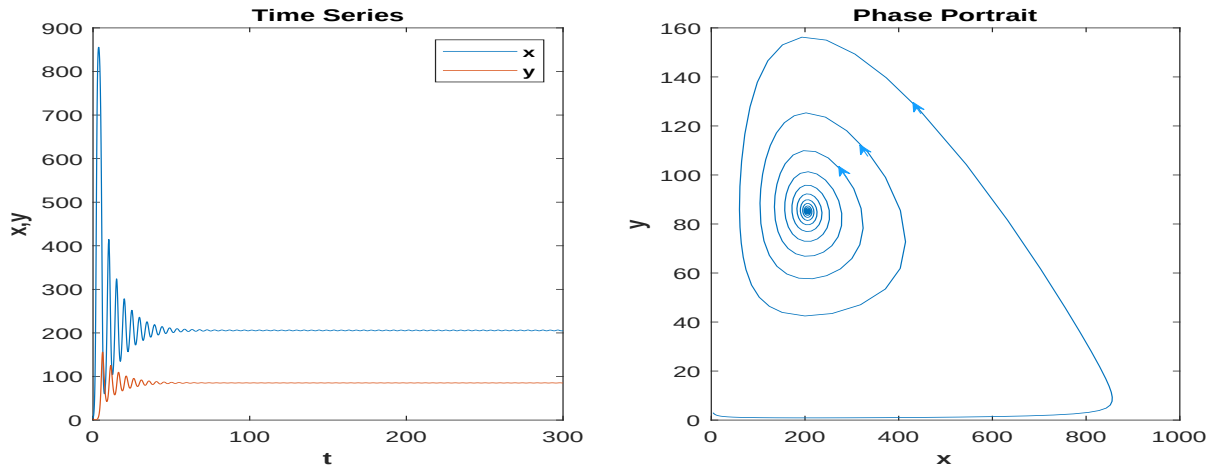


Figure: Behaviour of the dynamical system for the assigned values of the parameters (Arrows indicating the stability nature of the equilibrium point)

At $m = 0.1227$, *Hopf Bifurcation* occurs in the dynamical system, giving birth to a limit cycle. The corresponding phase portrait and time series of the system are as shown:

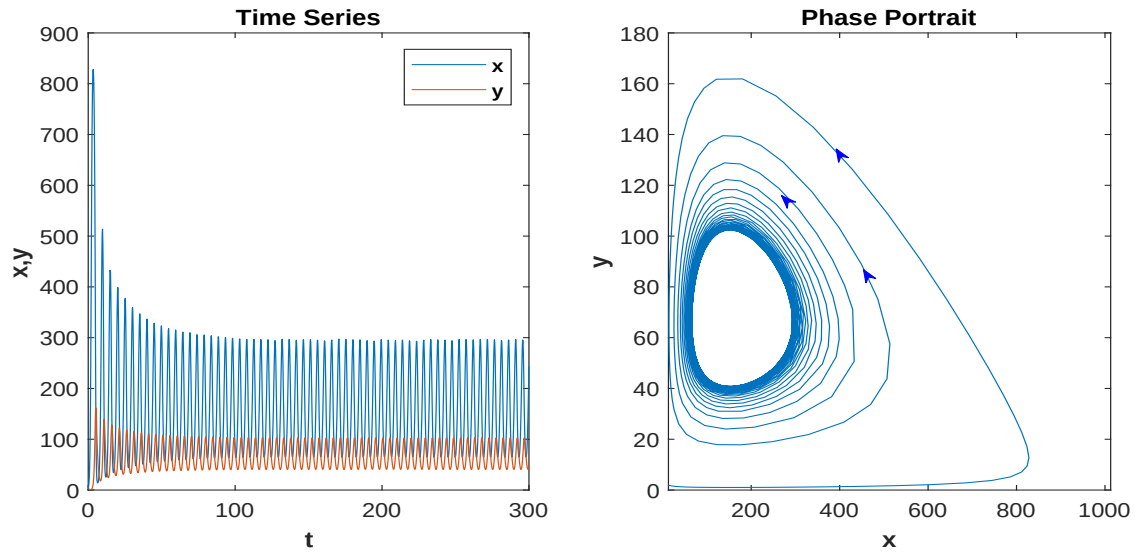


Figure: Change in behaviour of the dynamical system for the assigned values of the parameters (Arrows indicating the stability nature of the limit cycle)

For $m > 0.8445$, the equilibrium point E_2 is not feasible and for the higher values of m in the desired range, we see the nature of the equilibrium point E_1 . The corresponding phase portrait and time series of the system are as shown:

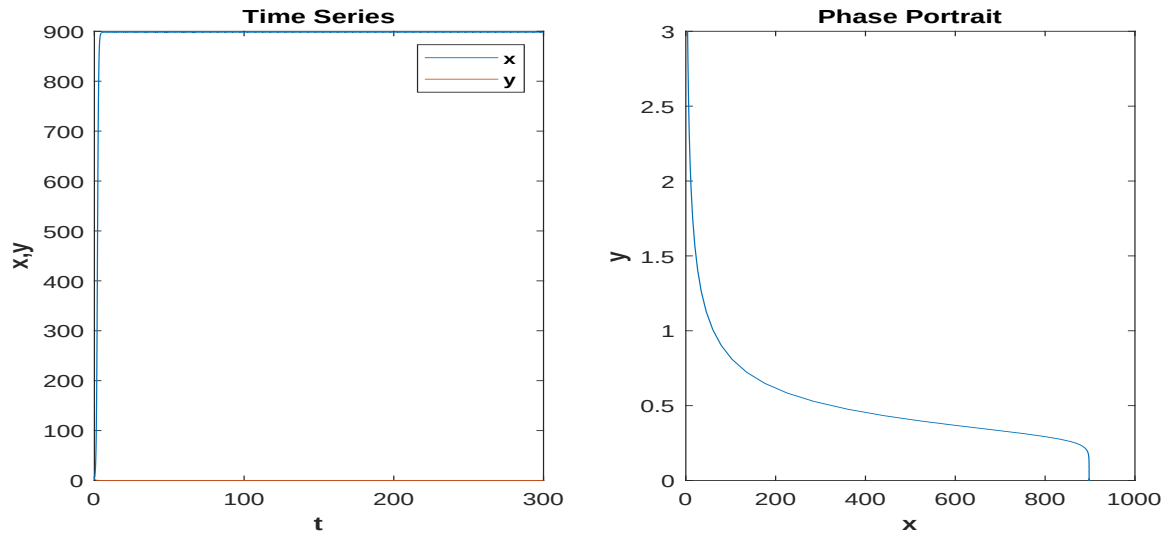


Figure: Behaviour of the point E_1 for higher values of m in the desired range

Chapter 6

Conclusion & Suggestion of Research Project

Conclusion:

The project work on Nonlinear Dynamical Systems has provided basic pieces of information about the behaviour of dynamical systems and the analysis regarding them. From studying the Nonlinear differential equations to the basic idea of chaos theory and linearization of the functions through Taylor Expansion, the project has significant impact on the area of Nonlinear Dynamics. From Stability analysis, the behaviour of the system on small perturbations and their reply can be verified. Either they can gain their stability back or move on from their stability position. From Bifurcation analysis, we know how a parameter can change the behaviour of a dynamical system through a series of changes in its value. The Chaos theory shows how the initial conditions affect the system's dynamics. Besides continuous dynamical systems, there are also discrete dynamical systems which are also of very much significance in so many fields. Also, by certain numerical techniques, we observed the change of dynamics in a system practically. Overall depending on the fundamental aspects of Dynamical Systems, one can go into deeper insights which are the main aspects of this project work.

Suggestion:

The project work on the topic of Nonlinear Dynamics demands some suggestions regarding its domain. The project work is done on one, two dimensional systems. Higher dimensional systems are not considered here. But in reality, higher dimensional systems has also a significant impact on different fields like Physics, Biology, Engineering and Economics. The stability analysis is done on the basis of linearized terms. The impact of borderline cases and thus of the higher-order terms in the Taylor Expansion series is not much discussed here but it actually plays a very important role in several cases. The discrete dynamical systems leading to the study of chaos must be thoroughly discussed. The computational techniques involved here are not that much of advanced level. With advanced computational techniques, better predictions can be made regarding the dynamical systems and their behaviours. Overall, the project has a significant impact on the fundamental aspects of Nonlinear Dynamical Systems. With advancements in technologies and theoretical analysis, better research work can be done in this discipline.

Chapter 7

Results Achieved

Continuous Dynamical Systems:

Consider the two dimensional dynamical system,

$$\begin{aligned}\frac{dx}{dt} &= rx(1 - \frac{x}{k}) - \frac{\alpha(1-m)xy}{1+\alpha h(1-m)x} \\ \frac{dy}{dt} &= \frac{\theta\alpha(1-m)xy}{1+\alpha h(1-m)x} - \delta y\end{aligned}$$

where $r, k, \alpha, m, h, \theta, \delta$ are the parameters (all positive) and $0 < m < 1$. In this system, for certain values of certain parameters, at $m = 1 - \frac{\theta+h\delta}{hk\alpha(\theta-\delta h)} = m^*$ (say) the equilibrium point changes from stable focus to unstable focus, thus giving birth to a limit cycle. And in this case, a limit cycle is born to balance the system dynamically. So a sudden change occurs in the dynamical system known as Bifurcation. For its unique characteristics in this case, it is known as Hopf Bifurcation.

In case of other bifurcations also, the system depends on certain parameters. Those cases are excluded in this chapter since they are discussed in previous chapters. Several cases are excluded in this topic since they are covered in the early chapters.

Discrete Dynamical Systems:

For Discrete Dynamical Systems, we work with iterated maps or difference equations. Consider the maps,

$$x_1 = f(x_0),$$

$$x_2 = f(x_1),$$

...

...

The sequence $x_0, x_1, \dots$ defines the trajectory of the variable x_n , where now the time n is discrete. In this case, we obtained that the condition of stability of x^* for the system is,

$$\left| \frac{df}{dx_n} \Big|_{x_n=x^*} \right| < 1$$

- **The Logistic Map:**

It is one of the most important concepts towards the development of Chaos theory. The logistic map is represented by the equation,

$$x_{n+1} = ax_n(1 - x_n) = f(x_n),$$

where a is a parameter (Intrinsic Growth Rate), $0 \leq a \leq 4$, $0 \leq x \leq 1$, $n = 0, 1, 2, \dots$

Equilibrium Point (x^*)	$\frac{df}{dx_n} _{x_n=x^*}$	Stability Condition
$x^* = 0$	a	$0 \leq a < 1$
$x^* = \frac{a-1}{a}$	$2 - a$	$1 < a < 3$

Table 3: Theoretical Analysis of the Logistic Map

Therefore from the above analysis,

(i) $x^* = 0$ is stable for $0 \leq a < 1$ and unstable for $a > 1$.

(ii) $x^* = \frac{(a-1)}{a}$ is stable for $1 < a < 3$ and unstable for any value outside this range.

Therefore, we can see that the point which is stable for $a < 1$ becomes unstable for $a > 1$ and the point which is unstable for $a < 1$ becomes stable for $a > 1$. So there is a change of stability in these points. Here a transcritical bifurcation occurs. Moreover for k-periodic solutions, much more complicated behaviour of the system is observed, called period-doubling bifurcations. The bifurcation diagram of this map is shown below:

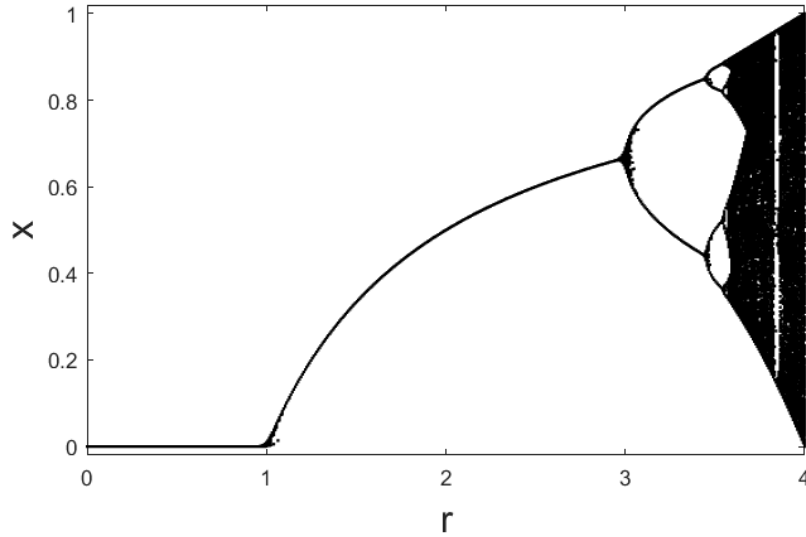


Figure: Bifurcation Diagrams of the Logistic Map for $a \in (0, 4)$

Chapter 8

References

- S.H. Strogatz: Nonlinear Dynamics and Chaos: With Applications in Physics, Biology, Chemistry and Engineering (Addison-Wesley, Manchester 1994).
- Layek, G. C. An introduction to dynamical systems and chaos. Vol. 449. New Delhi: Springer, 2015.
- Alligood, Kathleen T., Tim D. Sauer, James A. Yorke, and David Chillingworth. "Chaos: an introduction to dynamical systems." SIAM Review 40, no. 3 (1998): 732-732.
- Robinson, Rex Clark. An introduction to dynamical systems: continuous and discrete. Vol. 19. American Mathematical Soc., 2012.
- Bourgin, D. G. "Mathematics: Differential Equations. Shepley L. Ross. Blaisdell (Ginn), New York, 1964. xiv+ 594 pp. Illus. 10." Science 148.3674 (1965): 1213-1213.
- Lakshmanan, M., & Rajaseekar, S. (2012). Nonlinear dynamics: integrability, chaos and patterns. Springer Science & Business Media.
- Bairagi, N. and Jana, D., 2011. On the stability and Hopf bifurcation of a delay-induced predator–prey system with habitat complexity. Applied Mathematical Modelling, 35(7), pp.3255-3267.